Lab 1

x_{ij} - the value of the jth variable for the ith observation, $i=1,\dots,n$, $j=1,\dots,p$.

$$X=\begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}$$

$x_i=(x_{i1},x_{i2},\dots,x_{ip})^T$ - vector of length p, containing the p variable measurements for ith observation (vectors are represented as columns).

Variables are columns of X and are written as

$$x_j=\begin{bmatrix} x_{1j} \\ x_{2j} \\ \vdots \\ x_{nj} \end{bmatrix}$$

$$X=\begin{bmatrix} x_1 & x_2 & \cdots & x_p \end{bmatrix}=\begin{bmatrix} x_1^T \\ x_2^T \\ \vdots \\ x_n^T \end{bmatrix}$$

y_i - ith observation of the variable on which we wish to make predictions (i.e. response variable).

$$y=x_j=\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

Supervised Learning

Y - quantitative response,

$X_1,X_2,\dots,X_p$ - different predictors,

$X=(X_1,X_2,\dots,X_p)$,

We assume that there is some relationship between Y and X, which can be written in the form

$$Y = f(X) + \varepsilon,$$

where:

- f is some fixed but unknown function of $X_1, X_2, \ldots, X_p$.
- ε - random error term, which is independent of X and has mean zero ($E[\varepsilon]=0$).

Estimation of f

Prediction of Y using X:

$$\hat{Y} = \hat{f}(X),$$

where:

- $\hat{Y}$ - prediction of Y,
- $\hat{f}$ - estimation of f.

The accuracy of $\hat{Y}$ as a prediction for Y depends on two quantities:

- reducible error (which can be improved by more appropriate learning technique to estimate f),
- irreducible error (cannot be improved because Y is also a function of ε, which cannot be predicted using X).

$$E\left[(Y-\hat{Y})^2\right] = E\left[\left(f(X)+\varepsilon-\hat{f}(X)\right)^2\right] = \left(f(X)-\hat{f}(X)\right)^2 + Var(\varepsilon).$$

Parametric Methods

- We made an assumption about the functional form (or shape) of f.
- We estimate parameters on the basis of training data to fit or train the model.

Linear regression

$$f(X) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_p X_p$$

and we estimate $p+1$ parameters $\beta_0, \beta_1 \ldots, \beta_p$:

$$\hat{f}(X) = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2 + \ldots + \hat{\beta}_p X_p.$$

The most common approach to fit the model is referred to as ordinary least squares, and the commonly-used measure of quality of fit is the mean squared error:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{f}(x_i)\right)^2$$

where $\hat{f}(x_i)=\hat{y}_i$ is the prediction that $\hat{f}$ gives for the ith observation. But with no test observations we should use cross-validation method of estimating test MSE.

Simple Linear Regression

$$Y=\beta_0+\beta_1 X+\varepsilon .$$

Y - response variable,

X - predictor,

ε - random term with mean zero and variance σ^2 (stronger assumption $\varepsilon \sim N(0,\sigma^2)$.

β_0 - intercept term (the expected value of Y when $X=0$),

β_1 - slope (the average increase in Y associated with a one-unit increase in X),

$\bar{x}=\frac{1}{n}\sum_{i=1}^{n} x_i$ - sample average of X,

$\bar{y}=\frac{1}{n}\sum_{i=1}^{n} y_i$ - sample average of Y,

$s_x^2=\frac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})^2$ - sample variance of X,

$s_y^2=\frac{1}{n-1}\sum_{i=1}^{n}(y_i-\bar{y})^2$ - sample variance of Y,

$c_{xy}=\frac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})$ - sample covariance,

$r_{xy}=\frac{c_{xy}}{s_x s_y}=\frac{\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i-\bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i-\bar{y})^2}}$ - sample correlation.

Estimation of parameters β_0,β_1

- $\hat{y}_i=\hat{\beta}_0+\hat{\beta}_1 x_i$ - prediction for Y when $X=x_i$,
- $e_i=y_i-\hat{y}_i$ - ith residual,
- $RSS=e_1^2+e_2^2+\ldots+e_n^2$ - residual sum of squares.

$$RSS=RSS(\hat{\beta}_0,\hat{\beta}_1)=\sum_{i=1}^{n} e_i^2=\sum_{i=1}^{n}(y_i-\hat{\beta}_0-\hat{\beta}_1 x_i)^2 .$$

The least squares approach chooses $\hat{\beta}_0$ and $\hat{\beta}_1$ to minimize the RSS, and the solution is

$$\left\{\begin{array}{l}\hat{\beta}_1=\dfrac{\sum_{i=1}^{n}\left(x_i-\overline{x}\right)\left(y_i-\overline{y}\right)}{\sum_{i=1}^{n}\left(x_i-\overline{x}\right)^2},\\ \hat{\beta}_0=\overline{y}-\hat{\beta}_1\overline{x}.\end{array}\right\}$$

Theorem 1.

$\hat{\beta}_0,\hat{\beta}_1$ are unbiased estimators of β_0,β_1, i.e. $E\left[\hat{\beta}_0\right]=\beta_0$ and $E\left[\hat{\beta}_1\right]=\beta_1$. Moreover,

$$Var\left(\hat{\beta}_0\right)=\sigma^2\left(\frac{1}{n}+\frac{\overline{x}^2}{\sum_{i=1}^{n}\left(x_i-\overline{x}\right)^2}\right),Var\left(\hat{\beta}_1\right)=\frac{\sigma^2}{\sum_{i=1}^{n}\left(x_i-\overline{x}\right)^2}.$$

Assessing the Accuracy of the Model

Residual Standard Error (Estimation of σ)

The estimator of σ is a residual standard error:

$$RSE=\sqrt{\frac{RSS}{n-2}}.$$

R^2 *Statistic*

To calculate R^2, we use the formula

total sum square   residual sum square

$$R^2=\frac{TSS-RSS}{RSS},$$

where $TSS=\sum_{i=1}^{n}\left(y_i-\overline{y}\right)^2$ is the total sum of squares, which measures total variability of Y. So R^2 measures proportion of variability of Y that is explained by the model.

si r cuadrado es cercano a 1 es q es bueno, si es cercano a 0 es q no es bueno

Inference

Standard errors of estimators $\hat{\beta}_0,\hat{\beta}_1$:

$$SE\left(\hat{\beta}_0\right)=RSE\sqrt{\frac{1}{n}+\frac{\overline{x}^2}{\sum_{i=1}^{n}\left(x_i-\overline{x}\right)^2}},SE\left(\hat{\beta}_1\right)=RSE\sqrt{\frac{1}{\sum_{i=1}^{n}\left(x_i-\overline{x}\right)^2}}.$$

Theorem 2.

Under stronger assumption that $\varepsilon\sim N\left(0,\sigma^2\right)$, we have that

$$\frac{\hat{\beta}_0-\beta_0}{SE(\hat{\beta}_0)} \sim t_{n-2}, \frac{\hat{\beta}_1-\beta_1}{SE(\hat{\beta}_1)} \sim t_{n-2}.$$

Therefore, 95% confidence interval for β_1 is (approximately) of the form

$$\left[\hat{\beta}_1-2\,SE(\hat{\beta}_1), \hat{\beta}_1+2\,SE(\hat{\beta}_1)\right).$$

Precisely, $1-\alpha$ confidence interval for β_1 is of the form

$$\left[\hat{\beta}_1-t_{1-\alpha/2,n-2}SE(\hat{\beta}_1), \hat{\beta}_1+t_{1-\alpha/2,n-2}SE(\hat{\beta}_1)\right),$$

alfa suele ser 0,01 (pequeño). Intent9 buscar la prob de q el test static sea mayor, entonces la probabilidad tiene q ser pequeña

where $t_{1-\alpha/2,n-2}$ is the critical value, i.e. $P\left(|t_{n-2}|>t_{1-\alpha/2,n-2}\right)=\alpha$.

Standard errors can also be used to perform hypothesis tests on the coefficients. The most common hypothesis test involves testing the null hypothesis of

$$H_0: \text{There is no relationship between } X \text{ and } Y$$

versus the alternative hypothesis

$$H_1: \text{There is some relationship between } X \text{ and } Y$$

Mathematically, this corresponds to testing

$$H_0: \beta_1=0$$

obvio pq y=b0 + b1x

versus

$$H_1: \beta_1 \neq 0.$$

In practice, we compute a t-statistic, given by

$$t=\frac{\hat{\beta}_1-0}{SE(\hat{\beta}_1)},$$

queremos q haya relacion, si b1 es 0 NO HAY RELACION

which has Student t distribution with $n-2$ degrees of freedom. It measures the number of standard deviations that $\hat{\beta}_1$ is away from 0 (recall the 3 sigma rule or 68-95-99.7 rule). If $|t_0|$, (where t_0 equals value of statistic t from data) is large enough, we reject H_0. Simpler, we can compute the probability of observing any number equal to $|t_0|$ or larger in absolute value, assuming $\beta_1=0$:

$$P\left(|t| \geq |t_0|\right)=p\text{-value}.$$

We call this probability the p-value. A small p-value indicates that it is unlikely that $\beta_1=0$. The typical p-value cutoffs for rejecting the null hypothesis are 5% or 1% which corresponds to t-statistic of around 2 and 2.75, respectively.

Prediction

We have that:

$$Y=\beta_0+\beta_1 X+\varepsilon \vee X=x \Leftrightarrow Y(x)=\beta_0+\beta_1 x+\varepsilon(x)$$

$$Y(x)=\beta_0+\beta_1 x+\varepsilon(x) \vee E \Leftrightarrow \mu_{Y(x)}=E[Y(x)]=\beta_0+\beta_1 x+0$$

Hence we can estimate that:

$$\hat{\mu}_{Y(x)}=\hat{\beta}_0+\hat{\beta}_1 x$$

Since

$$Var(\hat{\mu}_{Y(x)}) \quad ¿Var(\hat{\beta}_0+\hat{\beta}_1 x)=Var(\overline{Y}-\hat{\beta}_1 \overline{x}+\hat{\beta}_1 x)=¿=\frac{\sigma^2}{n}+(x-\overline{x})^2\frac{\sigma^2}{\sum_{i=1}^{n}(x_i-\overline{x})^2}=\sigma^2\left(\frac{1}{n}+\frac{(x-\overline{x})^2}{\sum_{i=1}^{n}(x_i-\overline{x})^2}\right)$$

we can define

$$SE(\hat{\mu}_{Y(x)})=RSE\sqrt{\frac{1}{n}+\frac{(x-\overline{x})^2}{\sum_{i=1}^{n}(x_i-\overline{x})^2}}.$$

Theorem 3.

$$\frac{\hat{\mu}_{Y(x)}-\mu_{Y(x)}}{SE(\hat{\mu}_{Y(x)})} \sim t_{n-2}$$

The least square line:

$$\hat{Y}=\hat{\beta}_0+\hat{\beta}_1 X$$

is only an estimate for the true population regression line

$$f(X)=\beta_0+\beta_1 X$$

We can compute a confidence interval in order to determine how close $\hat{Y}$ will be to $f(X)$. This is the confidence interval for the average of $\hat{Y}$ when $X=x$ denoted by $\overline{y}(x)$.

But even if we knew $f(X)$ (we knew β_0, β_1) we cannot perfectly predict y because of the random error. The prediction interval is the measure how much will Y vary from $\hat{Y}$. This is the confidence interval for the $\hat{Y}$ when $X=x$ and then is much larger than the previous interval.

Since we have:

$$Y(x)=\beta_0+\beta_1 x+\varepsilon(x),$$

then we can estimate that:

$$\hat{Y}(x)=\hat{\beta}_0+\hat{\beta}_1 x+\varepsilon(x)=\hat{\mu}_{Y|x}+\varepsilon(x),$$

and

$$Var(\hat{Y}(x))=Var(\hat{\mu}_{Y|x})+\sigma^2=\sigma^2\left(\frac{1}{n}+\frac{(x-\bar{x})^2}{\sum_{i=1}^{n}(x_i-\bar{x})^2}\right)+\sigma^2.$$

So, we can define

$$SE(\hat{Y}(x))=RSE\sqrt{1+\frac{1}{n}+\frac{(x-\bar{x})^2}{\sum_{i=1}^{n}(x_i-\bar{x})^2}}.$$

Theorem 4.

$$\frac{\hat{Y}(x)-\hat{\mu}_{Y|x}}{SE(\hat{Y}(x))}\sim t_{n-2}.$$

Residues analysis

Since

$$Var(e_i)\ \text{¿}\ Var(Y_i-\hat{Y}_i)=Var(Y_i)+Var(\hat{Y}_i)-2Cov(Y_i,\hat{Y}_i)=\text{¿}=\sigma^2+\sigma^2\left(\frac{1}{n}+\frac{(x_i-\bar{x})^2}{\sum_{i=1}^{n}(x_i-\bar{x})^2}\right)-2Cov(Y_i,\hat{\beta}_0+\hat{\beta}_1 x_i)$$

we can define standard error

$$SE(e_i)=RSE\sqrt{1-\left(\frac{1}{n}+\frac{(x_i-\bar{x})^2}{\sum_{i=1}^{n}(x_i-\bar{x})^2}\right)}.$$

It can be shown that studentized residuals

$$r_i=\frac{e_i}{SE(e_i)}\sim t_{n-3}.$$

Exercice 1.

Consider the fitted values that result from performing linear regression without an intercept. In this setting, the ith fitted value takes the form

$$\hat{y}_i = x_i \hat{\beta},$$

where

$$\hat{\beta} = \frac{\sum_{j=1}^{n} x_j y_j}{\sum_{j=1}^{n} x_j^2}.$$

Show that we can write

$$\hat{y}_i = \sum_{j=1}^{n} a_j y_j.$$

What is a_j?

Note: We interpret this result by saying that the fitted values from linear regression are linear combinations of the response values.

Exercice 2.

In this problem we will investigate the t-statistic for the null hypothesis $H_0 : \beta = 0$ in simple linear regression without an intercept. To begin, we generate a predictor x and a response y as follows.

```
set.seed(1)
x <- rnorm (100)
y <- 2 * x + rnorm (100)
```

(a) Perform a simple linear regression of y onto x, without an intercept. Report the coefficient estimate β, the standard error of this coefficient estimate, and the t-statistic and p-value associated with the null hypothesis $H_0 : \beta = 0$. Comment on these results. (You can perform regression without an intercept using the command `lm(y~x+0)`).

(b) Now perform a simple linear regression of x onto y without an intercept, and report the coefficient estimate, its standard error, and the corresponding t-statistic and p-values associated with the null hypothesis $H_0 : \beta = 0$. Comment on these results.

(c) What is the relationship between the results obtained in (a) and (b)?

(d) For the regression of Y onto X without an intercept, the t-statistic for $H_0 : \beta = 0$ takes the form $t = \hat{\beta} / SE(\hat{\beta})$, where

$$SE(\hat{\beta}) = \sqrt{\frac{\sum_{i=1}^{n} (y_i - \hat{\beta} x_i)^2}{(n-1) \sum_{i=1}^{n} x_i^2}}$$

These formulas are slightly different from those given for simple regression, since here we are performing regression without an intercept.) Show algebraically, and confirm numerically in R, that the t-statistic can be written as

$$t = \frac{\sqrt{n-1}\sum_{i=1}^{n} x_i y_i}{\sqrt{\left(\sum_{i=1}^{n} x_i^2\right)\left(\sum_{i=1}^{n} y_i^2\right) - \left(\sum_{i=1}^{n} x_i y_i\right)^2}}.$$

(e) Using the results from (d), argue that the t-statistic for the regression of y onto x is the same as the t-statistic for the regression of x onto y.

(f) In R, show that when regression is performed with an intercept, the t-statistic for $H_0 : \beta_1 = 0$ is the same for the regression of y onto x as it is for the regression of x onto y.